

Breathing and Exchange of Gases - 100 MCQs

NCERT Nichod : Biology Practice Series NEET 2025



Good evening

↓
available?

↓
Ready



Breathing and Exchnage of Gases

100 microns



BIOLOGYLIVE

1. Which of the following structures close the glottis during swallowing to prevent the entry of food into wind pipe.

(1) Tongue

✓ (2) Epiglottis

(3) Diaphragm

(4) Larynx

→ elastic cartilaga. flap



2. Match Column - I with Column - II and select the correct option from the codes given below :

Column – I (Animals)

Column – II (Respiratory structures)

(A) Spiders

(B) King crab

(C) Prawn

(D) Labeo

(i) Gills

(ii) Book lungs

(iii) Books gills

(1) A-(iii), B-(i), C-(ii), D-(ii)

(3) A-(i), B-(ii), C-(iii), D-(ii)

✓ (2) A-(ii), B-(iii), C-(i), D-(i)

(4) A-(i), B-(iii), C-(i), D-(ii)



3. Lungs are enclosed in :

(1) Perichondrium

(2) Pericardium $\rightarrow$ Heart

✓ (3) Pleural membrane

(4) Peritoneum



4. Vital capacity of lungs is : 2025

(1) IRV + ERV

(2) IRV + ERV + TV – RV

(3) IRV + ERV + TV + RV x

(4) IRV + ERV + TV



5. Which of the following changes occur in diaphragm and intercostal muscles when expiration of air takes place ?

- (1) External intercostal muscles relax and diaphragm contracts
- (2) External intercostal muscles contract and diaphragm relaxes
- ✓ (3) External intercostal muscles and diaphragm relax .
- (4) External intercostal muscles and diaphragm contract



2025

6. Match column - I with column - II and select the correct option from the codes given below :

Column - I

A. $TV + ERV$

B. $RV + ERV + TV + IRV$

C. $ERV + RV$

Column - II

(i) Expiratory capacity

(ii) Total lung capacity

(iii) Functional residual capacity

✓ (1) A - (i), B - (ii), C - (iii)

(3) A - (iii), B - (ii), C - (i)

(2) A - (iii), B - (i), C - (ii)

(4) A - (ii), B - (iii), C - (i)



7. Match the column - I with column - II and select the correct option from the codes given below :

Column - I

- A. Tidal volume
- B. Inspiratory reserve volume
- C. Expiratory reserve volume
- D. Residual volume
- E. Vital capacity

Column - II

- (i) 2500 - 3000 mL of air
- (ii) 1000 mL of air
- (iii) 500 mL of air
- (iv) 4000 - 4600 mL of air
- (v) 1100 - 1200 mL – of air

(1) A-(iii), B-(iv), C-(ii), D-(i), E-(v)

(3) A-(iii), B-(i), C-(iv), D-(v), E-(ii)

✓ (2) A-(iii), B-(i), C-(ii), D-(v), E-(iv)

(4) A-(v), B-(i), C-(ii), D-(iii), E-(iv)



8. During expiration the diaphragm becomes :

- ✓✓ (1) Dome - shaped
- (2) Oblique
- (3) Normal
- (4) Flattened



9. The inspiratory reserve volume + tidal volume + expiratory reserve volume is the same as :

$$\overbrace{I.R.V + E.R.V}^{1.5}$$

2021

- ✓ (1) Inspiratory capacity + expiratory reserve volume
- (2) Total lung capacity – functional residual capacity
- (3) Inspiratory capacity + functional residual capacity
- (4) Inspiratory capacity + residual volume

$$I.R.V + T.V = \underline{1.5}$$



10. Which of the following is true for CO₂ oncentration?

- ✓ (1) More in alveolar air than in expired air
- (2) More in expired air than in alveolar air
- (3) More in inspired air than in alveolar air
- (4) More in inspired air than in expired air



11. The enzyme that increases the reaction rate between CO_2 and H_2O in red blood cells is :

- ✓ (1) Carbonic anhydrase
- (2) Adenylate cyclase
- (3) Carbonic synthetase
- (4) Alkaline phosphatase



12. The partial pressure of oxygen is maximum in :

(1) Alveolar air $\rightarrow$ 104 mmHg

(2) Arterial blood

(3) Venous blood

☒ (4) Expired air



13. Pneumotaxic centre which can moderate the functions of the respiratory rhythm centre is present at :

- ✓ (1) Pons region of brain
- (2) Thalamus
- (3) Spinal cord
- (4) Right cerebral hemisphere



14. Match the column - I with column - II and select the correct option from the codes given below :

Column - I

(A) Fish

(B) Biologically usefull energy

(C) 104 mm Hg

(D) Vocal cords

Column - II

(i) Larynx

(ii) Alveolar air P_{O_2}

(iii) Branchial respiration

(iv) ATP

✓ (1) A-(iii), B-(iv), C-(ii), D-(i)

(3) A-(iii), B-(iv), C-(i), D-(ii)

(2) A-(iv), B-(ii), C-(iii), D-(i)

(4) A-(i), B-(ii), C-(iv), D-(iii)



15. Emphysema is a condition resulting from :

- ✓ (1) Cigarette smoking ↻
- (2) Liquor consumption
- (3) Drug addiction
- (4) Reduced oxygen carrying capacity of blood



16. How many matching is/are correct

- Gill filaments – Fish ✓
- Cutaneous respiration – Earthworm ✓
- Diaphragm – Inspiration ✓

(1) One

(2) Two

✓ (3) Three

(4) None of these



17. It is known that exposure to carbon monoxide is harmful to animals because :-

- (1) It reduces CO_2 transport
- ✓ (2) It reduces O_2 transport ✓
- (3) It increases CO_2 transport
- (4) It increases O_2 transport

4b → CO



18. Mark the true statement among the following with reference to normal breathing

- (1) Inspiration is a passive process where as expiration is active
- ✓ (2) Inspiration is a active process where as expiration is passive
- (3) Inspiration and expiration are active processes
- (4) inspiration and expiration are passive processes



19. Mark the incorrect statement in context to O₂ binding to Hb.

- ✓ (1) ^{low water} Higher H⁺ conc.
- (2) Lower temperature
- (3) Lower pCO₂
- (4) Higher pO₂

↓
In alveolic



11.V.1m7
20. Respiratory process is regulated by certain specialized centres in the brain. One of the following listed centres can reduce the inspiratory duration upon stimulation

(1) Medullary inspiratory centre

✓ (2) Pneumotaxic centre

(3) Apneustic centre

(4) Chemosensitive centre



21. CO_2 dissociate from carbamino haemoglobin when :

(1) pCO_2 is high & pO_2 is low

✓✓ (2) pO_2 is high and pCO_2 is low = alveolic

(3) pCO_2 and pO_2 are equal

(4) None of the above



22. Match the following and mark the correct options

Animal

Respiratory Organ

A. Earthworm —————> (i) Moist cuticle

B. Aquatic Arthropods —————> (ii) Gills

C. Fishes —————> (iii) Lungs

D. Birds/Reptiles —————> (iv) Trachea

(1) A-(ii), B-(i), C-(iv), D-(iii)

(2) A-(i), B-(iv), C-(ii), D-(iii)

(3) A-(i), B-(iii), C-(ii), D-(iv)

✓ (4) A-(i), B-(ii), C-(ii), D-(iii)



23. Pharynx opens through the larynx region into

(1) nasal chamber

✓ (2) trachea

(3) sound box

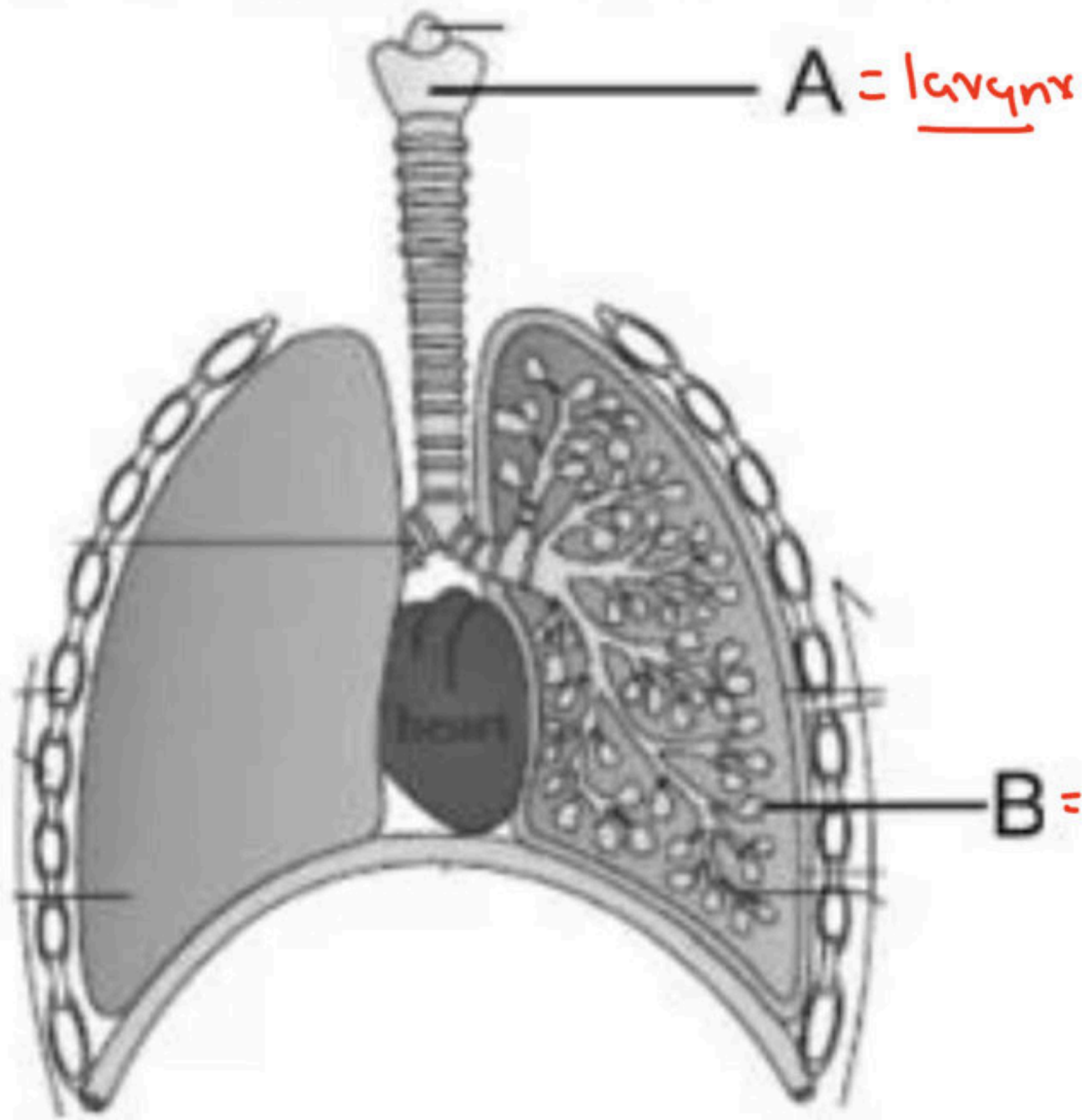
(4) none of these



24. In the given diagram A and B are

- (1) trachea, alveoli
- (2) larynx, bronchiole
- ☒ (3) larynx, alveoli
- (4) none of these





25. Which of the following is not true Thoracic chamber is made :-

- (1) dorsally by vertebral column
- (2) ventrally by sternum
- (3) laterally by ribs
- (4) lower side by ^{Dome}~~bean~~ shaped diaphragm

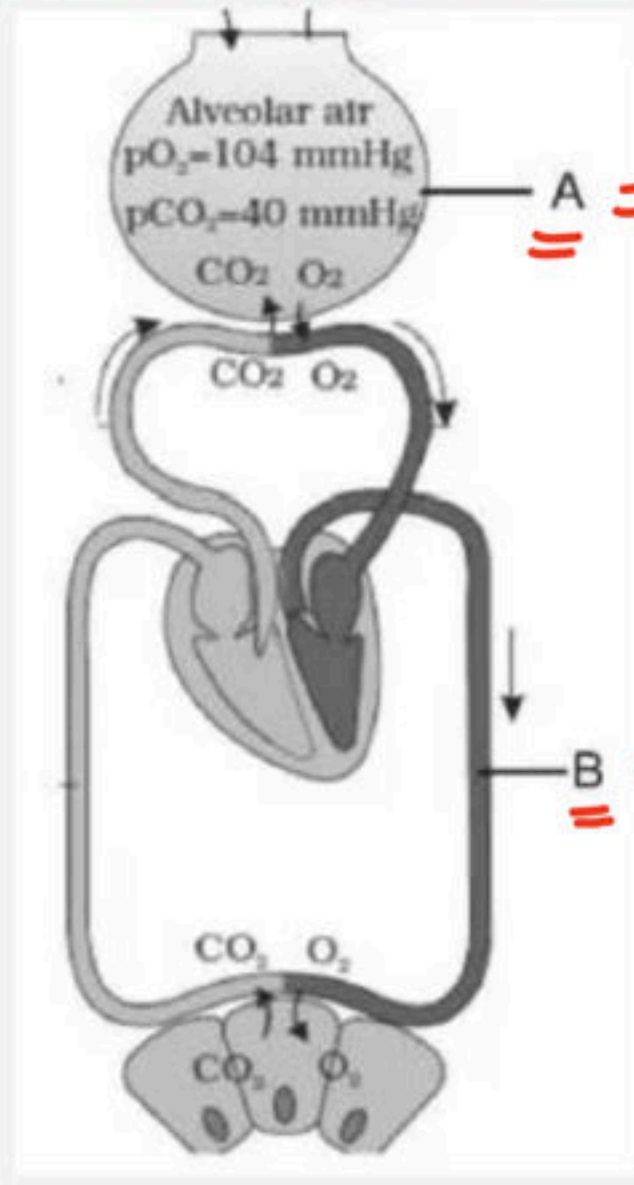


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26. During inspiration intrapulmonary pressure is decreased due to — — — —
— in pulmonary volume ^{↑SS}

- ✓ (1) increase
- (2) decrease
- (3) either 1 or 2
- (4) none of these



27. In the given diagram A and B are



A = alveolus

B = systemic A.

- (1) alveolus, systemic vein
- ✓ (2) alveolus, systemic artery
- (3) body tissues, systemic vein
- (4) none of these



28. Which of the following match is wrong regarding transport of gases during respiration

(1) 97% O₂, 20-25% CO₂

(2) 3% O₂

(3) 70% CO₂

✓ (4) ~~3%~~ CO₂

7%

RBCs

dissolved in plasma

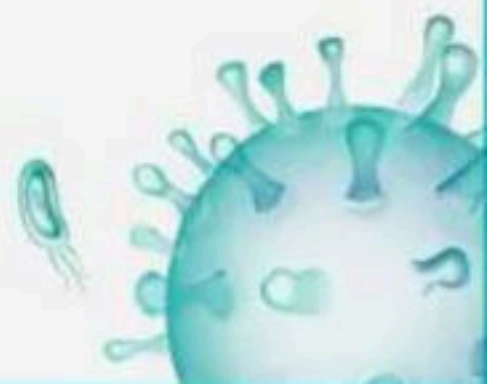
bicarbonate

dissolved in plasma



29. Binding of O₂ to haemoglobin depends upon

- (1) pO₂ and pCO₂
- (2) hydrgen ion concentration
- (3) temperature
- (4) all of these



30. Inspiration can occur if the intra-pulmonary pressure is :

- ✓ (1) Less than the atmospheric pressure = -ve pressure
- (2) More than the atmospheric pressure
- (3) Equal to the atmospheric pressure
- (4) Positive pressure in the lungs with respect to atmospheric pressure.



31

121. Every 100 mL of deoxygenated blood delivers —
mL of CO_2 to alveoli

(1) 10

(2) 20

✓ (3) 4

(4) none of these



v. imp

32. By the contraction of which one, the volume of the thoracic chamber increase in the anteroposterior axis:

- ✓ (1) Contraction of diaphragm ←
- (2) Contraction of lungs
- (3) Contraction of sternum
- (4) Contraction of external inter-costal muscle



33. Volume of air inspired or expired during a normal respiration is called :

- (1) Residual Volume
- (2) Expiratory Reserve Volume
- (3) Inspiratory Reserve Volume
- ✓ (4) Tidal Volume = 500ml



34. Neural signal from pneumotaxic centre can Reduce the duration of Inspiration and thereby alter the respiratory rate

- (1) reduce, expiration
- ✓ (2) reduce, inspiration
- (3) increase, expiration
- (4) increase, inspiration



35. Which of the following can sense changes in CO₂ and H⁺ concentration and send necessary signals to the rhythm centre

- (1) chemosensitive area ✓
- (2) receptors of aortic arch ✓
- (3) receptors of carotid artery ✓
- ✓ (4) all of these



36. Long exposure of dust in industry results in

- (1) lung inflammation ✓
- (2) fibrosis ✓
- (3) lung damage ✓
- (4) all of these ✓



37. Inspiration is initiated by :

- ✓ (1) Contraction of diaphragm
- (2) Contraction of lungs
- (3) Contraction of sternum
- (4) Contraction of internal inter-costal muscles



38. On an average, a healthy human breathes :

(1) 70-75 times/minute

(2) 5-10 times/minute

✓ (3) 12-16 times/minute

(4) 60-70 times/minute



39. Which is a inspiratory reserve volume :

(1) 1000 mL to 1100 mL

(2) 1100 mL to 1200 mL

(3) 500 mL

(4) None of these

↓
2500-3000ml



40. Which statement is correct for residual volume : 2028

(1) Volume of air inspired or expired during a normal respiration

(2) Additional volume of air, a person can inspire by a forcible inspiration

(3) Additional volume of air, a person can expire by a forcible expiration

✓ (4) Volume of air remaining in the lungs even after a forcible expiration → 1100-1200ml



41. Volume of air that will remain in the lungs after a normal expiration called :

202 S

- (1) Inspiratory Capacity
- (2) Expiratory Capacity
- ✓ (3) Functional Residual Capacity
- (4) Vital Capacity



42. Which statement is correct :

(1) Total Lung Capacity includes RV, ERV, TV, IRV ✗

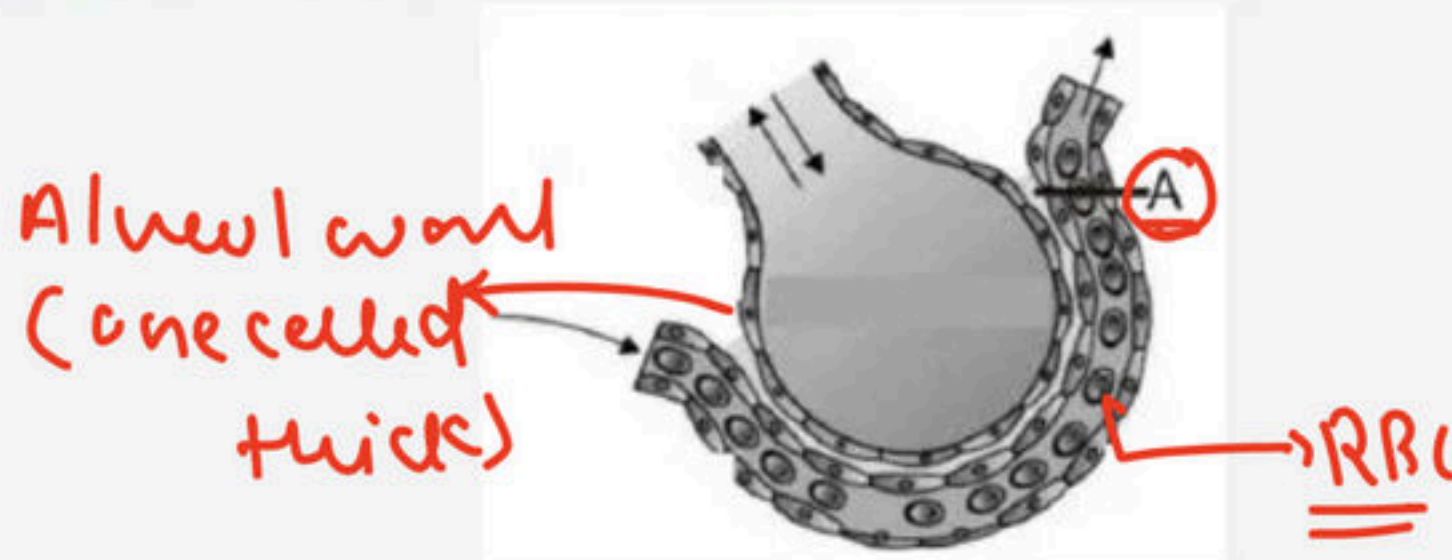
(2) Functional Residual Capacity includes ERV, TV RV ✗

✓ (3) Vital Capacity includes ERV, TV and IRV ✓

(4) Expiratory Capacity includes TV and ERV



43. Identify A in given diagram of an alveolus with a pulmonary capillary :



- (1) Mesothelium
- (2) Red blood cell
- (3) Alveolar wall
- (4) Basement substance

✓



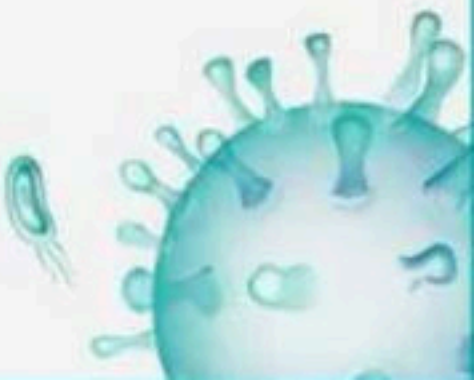
44. Which percent of O₂ is transported by RBCs :

- ✓ (1) 97 % ✓
- (2) 87 %
- (3) 70 %
- (4) 3 %



45. Which factor is favourable for the formation of oxyhaemoglobin :

- ✓ (1) High pO₂
- (2) High pCO₂
- (3) Low pH
- (4) None of these



46. Additional volume of air, a person can inspire by a forcible inspiration is called :

- (1) Residual Volume
- (2) Expiratory Reserve Volume
- ✓ (3) Inspiratory Reserve Volume
- (4) Tidal Volume



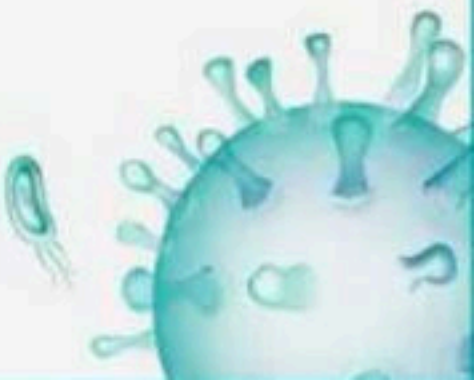
47. Rate and depth of respiration shall increase when :-

- (1) O₂ conc. increases
- ✓ (2) CO₂ conc. increases ↑
- (3) Haemoglobin increases
- (4) Vitamin-B decreases



48. A chemosensitive area situated adjacent to the rhythm center, is highly sensitive to

- (1) CO_2
- (2) H^+
- (3) O_2
- (4) Both A and B



49. On mountain climbing respiration becomes difficult due to less O_2 available. This condition is compensated by the body with this mechanism :

- ✓✓ (1) Increasing erythropoiesis →
- (2) Increasing fat deposition
- (3) Decreasing stool flow
- (4) None

C.B.Q.



50. Which volume has the maximum magnitude :

(1) Tidal volume $\times$

(2) E.R.V.

✓ (3) I.R.V. = 2500 - 3000 ml

(4) R.V.



51. Vital capacity is measured with :

- (1) Stethoscope
- ✓ (2) Spirometer
- (3) Aspirator
- (4) Sphygmomanometer



52. Which is a correct statement

- a. External intercostal muscles have no role in inspiration ✗
- b. Normal breathing rate is 12-16/minute ✓
- c. Pulmonary function can be assessed by spirometry ✓
- d. Tidal volume is 600 ml ✗

(1) a & b ~~(2)~~ b & c

(3) c & d (4) a & d



53. In normal respiration, which has no role :

- (1) Intercostal muscles
- (2) Diaphragm
- (3) Sternum
- (4) Spleen



54. Additional volume of air, a person can expire by
a forcible expiration is :

(1) IRV

✓ (2) ERV

(3) TV

(4) RV



55. What is the value of residual volume in a adult healthy person :

(1) 500 ml

(2) 1000 ml $\rightarrow$ E.R.V

☒ (3) 1200 ml

(4) None



56. Special vascularised structures called gills are used by most of the aquatic arthropods and molluscs while vascularized bags called lungs are used by the

- (1) Aquatic animal
- ✓ (2) Terrestrial animal
- (3) Protozoan and ascaris
- (4) All of these



57. Which is a correct statement

- (a) Trachea is a straight tube extending upto the mid abdominal cavity ^{Thoracic} ✗
- (b) Pleural fluid is present between two lungs pleura ✗
- (c) Thoracic chamber is dorsally formed by cervical vertebrae ✗
- (d) We can not directly alter the pulmonary volume

(1) a and b

(2) b and c

(3) only c

✓ (4) only d



58. Solubility of the gases as well as the thickness of the membranes involved are some important factors that can affect

(1) Rate of active transport

✓ (2) Rate of diffusion

(3) Rate of imbibition

(4) All of these



59. Which statement is true ∴

- (a) In blood CO₂ compratively diffuse easily because of its high solubility ✓
- (b) Approximately ~~18~~⁷~~19~~% CO₂ is transported by plasma ✗
- (c) CO₂ diffuses into blood passes into RBCs and reacts with H₂O to form H₂CO₃ ✓
- (d) Chloride ions diffuse from plasma into RBCs to maintain ionic balance ✓

(1) a, b and c ✓(2) a, c and d

(3) b, c and d (4) b and d



60. Which is not a true statement

- (a) $p\text{CO}_2$ of deoxygenated blood is ~~95~~⁴⁵ mm Hg
- (b) $p\text{CO}_2$ of alveolar air is 40 mm Hg ✓
- (c) $p\text{O}_2$ of alveolar air is 104 mm Hg ✓
- (d) $p\text{CO}_2$ of deoxygenated blood is 45 mm Hg ✓

(1) a and b

(2) b and c

(3) only c

✓ (4) only a

30 sec



61. Match Column - I with Column - II and select the correct option from the codes given below :

Column - I

(Animals)

(A) Pigeon

(B) Scorpion

(C) Planaria

(D) Earthworm

Column - II

(Respiratory structures)

(i) Book lungs

(ii) Skin

(iii) Lungs

(iv) Body surface

✓ (1) A-(iii), B-(i), C-(iv), D-(ii)

(3) A-(ii), B-(iv), C-(i), D-(iii)

(2) A-(i), B-(ii), C-(iii), D-(iv)

(4) A-(iii), B-(iv), C-(ii), D-(i)



62. A person is suffering from wheezing due to inflammation of bronchi and dyspnoea. His complaints increase in dusty environment. He may suffer from which disease

- (1) Pancreatitis
- (2) Jaundice
- ✓ (3) Asthma
- (4) Typhoid



63. Which is a correct statement

(a) Carbonic anhydrase is only present in RBC ✗

(b) O_2 is respiratory controlling power ✗

(c) Carbamino haemoglobin is a combination of Hb with CO ✗

(d) About 7% of CO_2 is carried by plasma ✓

(1) a and b (2) b and c

(3) only b ✓ (4) only d



64. Which is an incorrect statement ∴

(a) Spirometer has no clinical significance ✗

(b) A healthy man can inspire or expire approx-6000 to 8000 ml of air per second min ✗

(c) Expiratory reserve volume is 1000 ml ✓

(d) Normally, expiration is a passive process ✓

✓ (1) a and b

(2) b and c

(3) c and d

(4) a and d



65. Which is a correct matching

Column-I

(a) Opening of nasal chamber

(b) Respiration in adult frog

(c) Respiration in aquatic arthropod

(d) Membrane around lungs

Column-II

i. Pleura

ii. Gills

iii. Lungs

iv. Nasapharynx

✓ (1) a-iv, b-iii, c-ii, d-i

(2) a-iv, b-iii, c-i, d-ii

(3) a-iv, b-ii, c-i, d-iii

(4) a-i, b-ii, c-iii, d-iv



66. Match Column - I with Column - II and select the correct option from the codes given below :

Column - I

(A) Trachea

(B) Respiratory centre

(C) Yeast

(D) Insects

Column - II

(i) Ethanol

(ii) Medulla oblongata

(iii) Tracheal respiration

(iv) Cartilaginous rings

(1) A-(iii), B-(iv), C-(i), D-(ii)

(2) A-(iv), B-(i), C-(ii), D-(iii)

✓ (3) A-(iv), B-(ii), C-(i), D-(iii)

(4) A-(i), B-(iv), C-(ii), D-(iii)



67. Which is a correct matching set

500ml or 0.5L

Column-I

Column-II

a. pO_2 in the alveoli

b. pO_2 of oxygenated blood

c. ERV

d. Tidal volume

i. 0.5 L

ii. 1 L

iii. 95 mm Hg

iv. 104 mm Hg

~~(1)~~ a-iv, b-iii, c-ii, d-i

(2) a-i, b-iii, c-ii, d-iv

(3) a-iii, b-ii, c-iv, d-i

(4) a-ii, b-iii, c-iv, d-i



68. Match the column - I with column - II and select the correct option from the codes given below :

Column - I

A. Tracheoles

B. Carbonic anhydrase

C. Lactic acid

D. Fermentation

Column - II

(i) Insect

(ii) Bicarbonates

(iii) Fast muscle

(iv) Yeast

Exemplar

☒ (1) A-(i), B-(ii), C-(iii), D-(iv)

(3) A-(i), B-(ii), C-(iv), D-(iii)

(2) A-(iv), B-(iii), C-(ii), D-(i)

(4) A-(iii), B-(ii), C-(i), D-(iv)



69. Which one respire through gills :

(1) Crocodile

(2) Whale

(3) Frog

✓ (4) Prawn



70. Air is warmed as it passes through :

- ✓✓ (1) Nasal cavities
- (2) Trachea
- (3) Bronchial tree
- (4) Alveoli



71. In tissue $\overset{40}{\text{PO}_2}$ and $\overset{45}{\text{PCO}_2}$ are respectively:

2025

- (1) 45 mmHg and 40 mmHg
- ✓ (2) 40 mmHg and 45 mmHg
- (3) 40 mmHg and 40 mmHg
- (4) 45 mmHg and 45 mmHg



72. Arrange the following in order of increasing volume:

a. Tidal volume

b. Residual volume

c. Expiratory reserve volume

d. Vital capacity

(1) $a < b < c < d$ (2) $a < d < c < b$

(3) $a < c < b < d$ (4) $a < d < b < c$

2023

$$\begin{array}{c} \text{TV} \\ 500 \text{ ml} \end{array} < \begin{array}{c} \text{ERV} \\ 1200 \text{ ml} \end{array} < \begin{array}{c} \text{RV} \\ 1200 \text{ ml} \end{array}$$

4600 ml



73. Vital capacity of lungs is :

- (1) Inspiratory reserve volume plus expiratory reserve volume
- (2) Inspiratory reserve volume plus tidal volume
- (3) Total lung capacity minus expiratory reserve volume
- ✓ (4) Total lung capacity minus residual volume



$\rightarrow 1200\text{ml}$

74. Residual volume is :

 $\rightarrow 4600$

(1) Greater than vital capacity

✓ (2) Greater than tidal volume 500ml

(3) Lesser than tidal volume

(4) Greater than inspiratory reserve volume



75. What is true about RBCs in human :

- (1) They transport 99.5% of O_2 ✗
- (2) They transport 80% oxygen; the 20% being transported by plasma ✗
- (3) They do not carry CO_2 at all ✗
- ✓ (4) They carry 20 - 25% of CO_2



76. In which animal transport of atmospheric air by tracheal tubes :

V.V.I.M.P

- (1) Hydra
- ☒ (2) Cockroach
- (3) Sponges
- (4) Earthworm



77. Skin is an accessory organ of respiration in :

(1) Humans

☒ (2) Frog

(3) Rabbit

(4) Lizard



78. Which statement is incorrect :

- (1) Lower invertebrates like sponges, coelenterates, flatworms, etc., exchange O_2 with CO_2 by simple diffusion over their entire body surface.
- ✓ (2) Trachea is a straight tube extending up to the midthoracic cavity, which divides at the level of ~~7th~~^{5th} thoracic vertebra into a right and left primary bronchi.
- (3) A healthy man can inspire or expire approximately 6000 to 8000 mL of air per minute.
- (4) pO_2 is a major factor which could affect binding of CO_2 with haemoglobin.



79. Which statement is incorrect :

- (1) Special vascularised structures called gills are used by most of the aquatic arthropods and molluscs
- (2) The part starting with the external nostrils up to the terminal bronchioles constitute the conducting part
- ✓ (3) The thoracic chamber is formed ventrally by the ~~vertebral column~~^{Sternum}, dorsally by the sternum¹⁻⁵, laterally by the ribs and on the lower side by the domeshaped diaphragm
- (4) Inspiration can occur if the pressure within the lungs is less than the atmospheric pressure



80. Intercostal muscles occur in :

(1) Abdomen

(2) Thigh

✓ (3) Ribs

(4) Diaphragm



81. Which one is a incorrect matching :

- (1) Inspiratory Capacity–3500 mL
- (2) Expiratory Capacity–1600 mL
- (3) Functional Residual Capacity–2300 mL
- (4) Vital Capacity–~~3600~~ mL

4600



82. The exchange of gases in the alveoli of the lungs takes place by :

✓ (1) Simple diffusion

(2) Osmosis

(3) ~~Active transport~~ → Inspired

(4) ~~Passive transport~~ → Im

↓
expiration



83. In alveoli of the lungs, the air at the site of gas exchange, is separated from the blood by :

2025

- (1) Alveolar epithelium only
- ✓ (2) Alveolar epithelium and capillary endothelium
- (3) Alveolar epithelium, capillary endothelium and tunica adventitia
- (4) Alveolar epithelium, capillary endothelium, a thin layer of tunica media and tunica adventitia



84. Oxygen dissociation curve of haemoglobin is :

- ✓✓ (1) Sigmoid ↪
- (2) Hyperbolic
- (3) Linear
- (4) Hypobolic



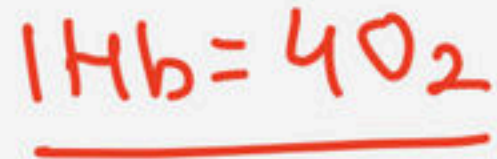
85. A molecule of haemoglobin carries oxygen
molecules:

(1) 1

(2) 2

(3) 3

✓ (4) 4



86. Expiratory capacity can be represented as

(1) V.C. + R.V.

(2) R.V. + E.R.V. + T.V. + I.R.V.

✓ (3) T.V. + E.R.V.

(4) Both A and B



87. Which volume has the minimum magnitude :

- ✓✓ (1) Tidal volume
- (2) E.R.V.
- (3) I.R.V.
- (4) R.V.



88. In sponge exchange of O₂ with CO₂ is done by :

- (1) Lungs
- (2) Gills
- (3) Trachea
- (4) Entire body surface



89. In insect respiration done by :

(1) Lungs

(2) Gills

✓ (3) Trachea ✓

(4) Entire body surface



90. The part starting with the external nostril up to the terminal bronchioles called :

- ✓ (1) Conducting part
- (2) Exchange part → ✗
- (3) Respiratory part
- (4) Both 2 & 3



91. The partial pressure of oxygen is equal in

- (1) Atmospheric air and Alveoli
- (2) Alveoli and Oxygenated blood
- (3) Alveoli and Deoxygenated blood
- (4) Deoxygenated blood and Tissues

$p_{CO_2} \rightarrow$ Alveoli $\rightarrow$ 40



92. CO_2 is carried by haemoglobin as:

- ✓ (1) carbamino-haemoglobin
- (2) carboxy-haemoglobin - rc
- (3) deoxy-haemoglobin
- (4) oxidised haemoglobin



93. Match the items given in Column I with those given in Column II and select the correct option given below:

Column I	Column II
i. Tidal Volume	a. 2500-3000 mL
ii. Inspiratory Reserve Volume	b. 1100-1200 mL
iii. Expiratory Reserve Volume	c. 500-550 mL
iv. Residual Volume	d. 1000-1100 mL

(1) i-a, ii-d, iii-c, iv-b

✓ (2) i-c, ii-a, iii-d, iv-b

(3) i-c, ii-b, iii-a, iv-d

(4) i-d, ii-c, iii-b, iv-a



94. Total volume of air accommodated in the lungs at the end of a forced inspiration is:

(1) $VC + FRC$

(2) $IC + ERV$

(3) $ERV + TV + IRV - EC$

✓ (4) $TV + IRV + \underline{\underline{FRC}}$



95. After a deep inspiration and maximum expiration, the capacity of lungs is known as

- ✓ (1) vital capacity
- (2) tidal volume
- (3) IRV
- (4) ERV



96. Consider the characters of a respiratory control center in humans:

I. It is located in pons

II. It moderates the functions of respiratory rhythm center

III. Its absence results in an increase in depth of respiration

The respiratory control center is:

- ✓ (1) Pneumotaxic center
- (2) Apneustic center
- (3) Chemosensitive zone
- (4) Dorsal group



97. Match List I with List II:

List I	List II
(1) Expiratory capacity	I. Expiratory reserve volume + Tidal volume + Inspiratory reserve volume
(2) Functional residual capacity	II. <u>Tidal volume</u> + <u>Expiratory reserve volume</u>
(3) Vital capacity	III. Tidal volume + Inspiratory reserve volume
(4) Inspiratory capacity	IV. <u>Expiratory reserve volume</u> + <u>Residual volume</u>

Choose the correct answer from the options given below:

(1) A-III, B-II, C-IV, D-I

(2) A-II, B-I, C-IV, D-III

(3) A-I, B-III, C-II, D-IV

☒ (4) A-II, B-IV, C-I, D-III



98. Trachea divides into two right and left primary bronchi at the level of:

- (1) Seventh cervical vertebra
- (2) Third thoracic vertebra
- ✓ (3) Fifth thoracic vertebra
- (4) Seventh thoracic vertebra



99. Consider the two statements:

Statement I: Expiratory Capacity is the total volume of air a person can expire after a normal inspiration. ✓

Statement II: Functional Residual Capacity is the volume of air that will remain in the lungs after a ^{normal} forceful maximal expiration. ✗

(1) Statement I is correct; Statement II is correct

✓(2) Statement I is correct; Statement II is incorrect

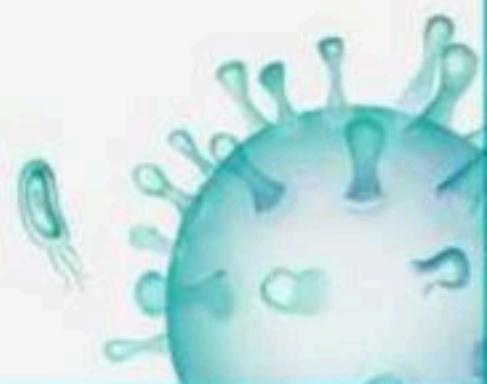
(3) Statement I is incorrect; Statement II is correct

(4) Statement I is incorrect; Statement II is incorrect



100. Chemosensitive area of respiratory centre medulla is affected by

- (1) less CO_2 and H^+ ions
- (2) less O_2 and H^+ ions
- ✓ (3) excess CO_2 and H^+ ions
- (4) excess O_2 and H^+ ions.



THANK YOU



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Thank You